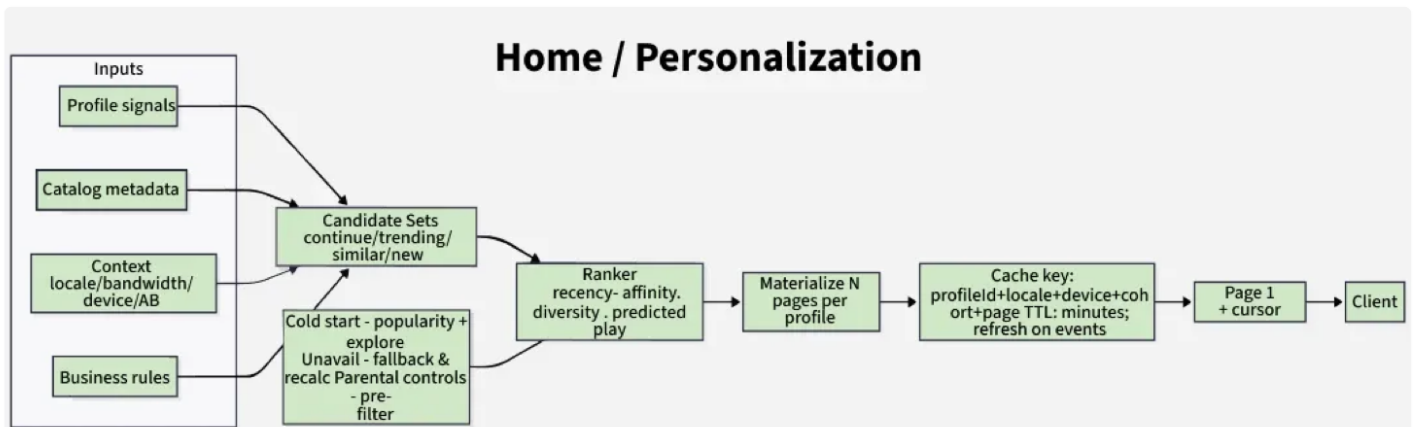


Use-Case Design (Product Surfaces)

* Home / Personalization

Show each profile a fast, relevant home page ("rows") that feels fresh But loads in ~100 - 300 ms P99 from cache



Inputs

- Profile signals: history (watched, abandoned), ratings/likes, search clicks, time-of-day/device.
- Catalog metadata: title, genre, maturity, language/availability, artwork variants.
- Context: locale, bandwidth class, device class (TV vs mobile), A/B cohort.
- Business rules: licensing windows, maturity controls, "continue watching" pinning.

Flow

1. Fetch candidate sets for the profile (continue watching, trending, similar-to-X, new releases).
2. Rank rows and within-row items using a scoring model (recency, affinity, diversity, predicted play).
3. Materialize N pages (e.g., 20–40 items per row) and cache per profile.
4. Return page 1 with a next-cursor; subsequent pages hydrate progressively.

Caching

- Key: profileId + locale + deviceClass + cohort + page.
- TTL: minutes; refresh on significant events (new watch, rating, time-bucket change).
- Negative cache brief 404s for removed titles to avoid stampedes.

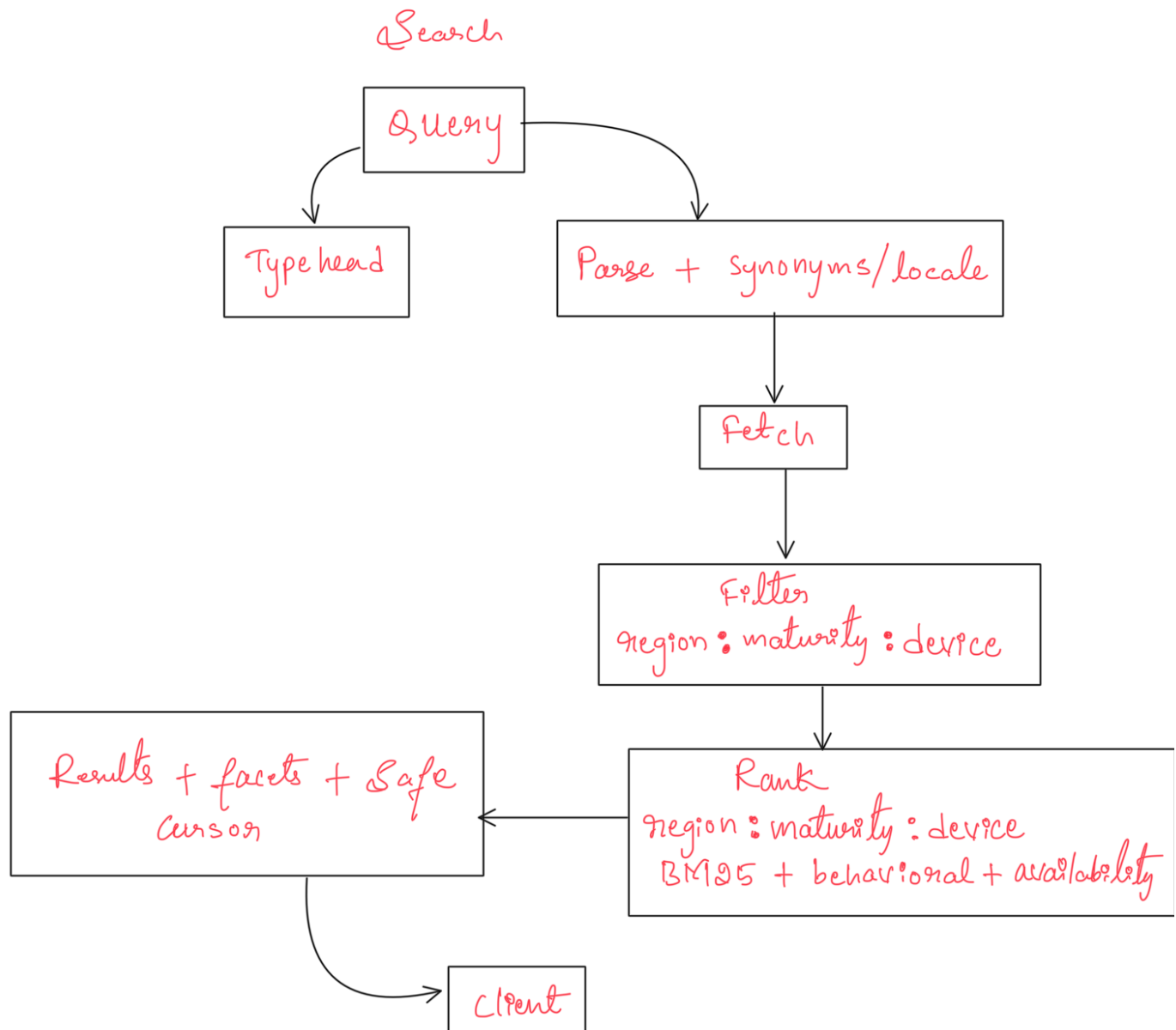
Edge cases

- New profile cold-start → popularity-based rows plus lightweight exploration.
- Title becomes unavailable mid-browse → even with fallback and mark for recalc

- Title becomes unavailable mid-browse → swap with fallback and mark for recall.
- Parental controls → filter candidates pre-rank.

Search

Instant, relevant findability with typeahead and robust filtering.



Inputs

- Query text, language, region; profile signals (preferred languages/genres).
- Inverted index (titles, people, collections), synonyms, spelling expansions.
- Behavioral features: prior clicks, completions, dwell.

Flow

1. Typeahead (prefix and fuzzy) returns entities: titles, people, genres, collections.
2. Full query parses tokens/hashtags, applies synonyms and locale rules.
3. Candidate fetch from index; filter by region, maturity, device capability (e.g., HDR).
4. Rank by a blend of lexical (BM25) + behavioral (affinity, popularity, recentness) + availability signals.
5. Return results with facets (genre, language, HDR, 4K) and safe pagination cursors.

Filters & facets

- Genre, language, audio/subtitle availability, resolution (HD/4K), HDR, release year.
- Personalization weight is lower than Home to respect query intent.

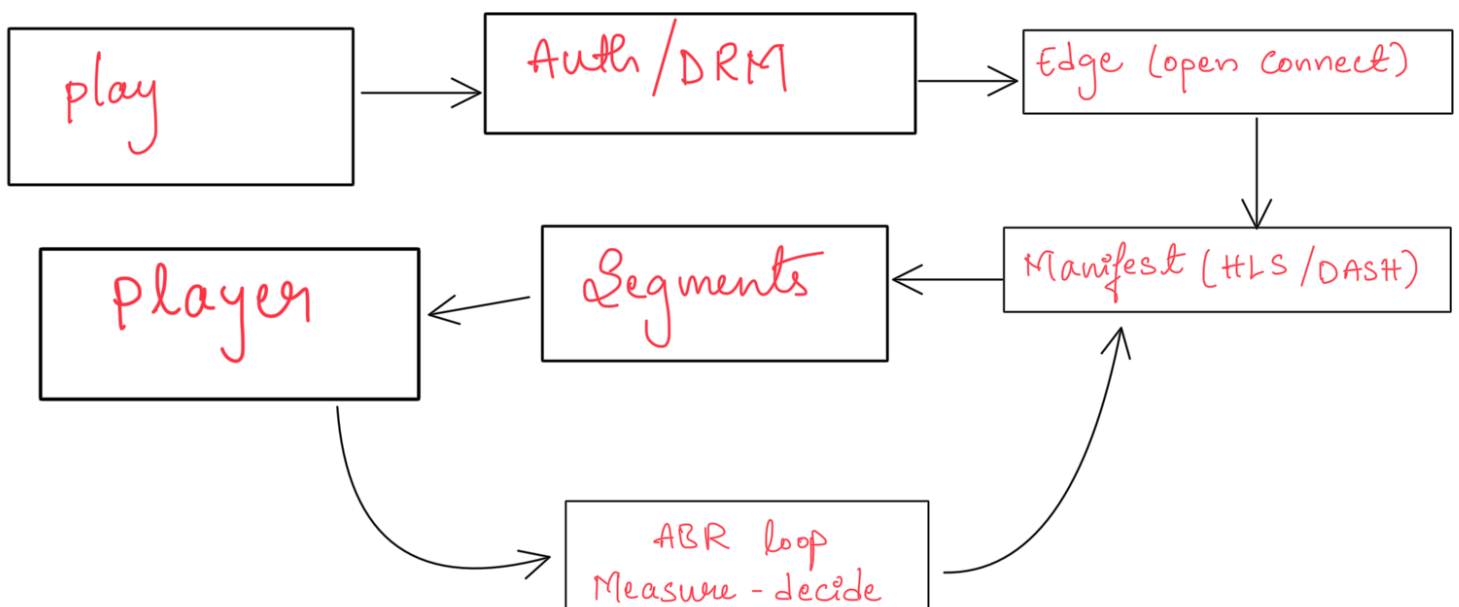
Edge cases

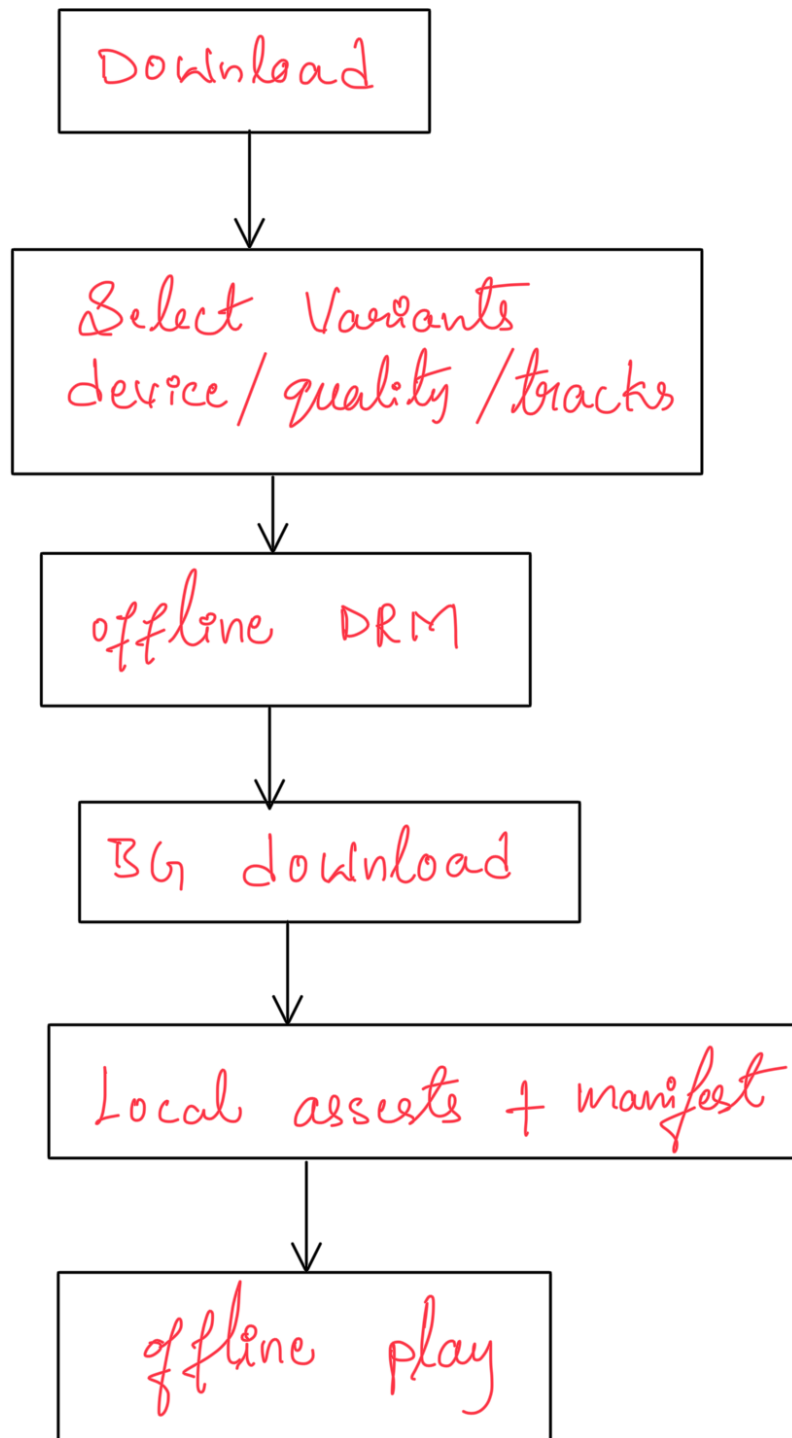
- Zero results: relax constraints (language → any), show “try these” suggestions.
- Ambiguous names (remakes): group by franchise/collection to reduce clutter.
- Regional gaps: show “notify me” or similar titles.

Playback

Quick start, Minimal rebuffers, Smooth quality ramps;
Enforce DRM and entitlements.

play Back & Downloads





Inputs

- Title/episode, profile entitlements, device capabilities (codec, DRM, max res), current bandwidth.
- Edge health (nearest Open Connect appliance), ABR ladder for the asset, captions/audio tracks.

Flow

1. Authorization: profile + entitlement check, license generation for DRM.
2. Edge selection: pick nearest healthy CDN node; return manifest (HLS/DASH) with track

variants.

3. ABR loop client-side: choose initial rung conservatively; monitor throughput/buffer; step up/down.
4. Telemetry streaming: startup milestones, bitrate switches, rebuffers, errors.

Tracks & features

- Multiple audio tracks, captions/subtitles; forced subtitles; accessibility tracks (AD).
- Trick-play thumbnails; preview scrubbing; instant resume from “continue watching”.

Error handling

- Edge failure → fast fallback to sibling edge or alternate region.
- License failure → retry with back-off; clear, actionable UI error if persistent.
- Segment 404 → skip to next segment; clamp ABR upward moves until stable.

Downloads (offline)

Reliable offline playback with correct rights and efficient storage/battery use.

Inputs

- Title availability for offline (licensing), device storage, network type, battery state, device constraints (codec/resolution).

Flow

1. Request download: compute eligible variants based on device profile and user choice (audio/subtitle, quality).
2. Obtain offline DRM license with validity window; bind to device/profile.
3. Download segments in background with throttling and network rules (Wi-Fi only, charging).
4. Maintain manifest of downloaded assets; validate license before play; auto-renew where allowed.

Space & lifecycle

- Size estimation shown before download; allow partial selections (episodes, tracks).
- Eviction policy: oldest unwatched, expired licenses, user-selected removal.
- Repackaging/pruning when codecs/bitrates change across app updates.

Edge cases

- Region change after travel: title plays if license remains valid; renewal may be blocked until in-region.
- Device clock drift: license checks use secure time sources.
- Multiple profiles on one device: enforce per-profile quotas and visibility.